COCA-COLA REFRESHMENTS EQUIPMENT BULLETIN

Service

DATE: July 9th, 2014	**NUMBER:** SERV – 312

SUBJECT: DN3800/5800 Glass Front Vender XY Mechanism Troubleshooting

SITUATION / BACKGROUND:

Crane Merchandising Systems (Dixie-Narco) DN3800 and DN5800 Glass Front Venders may sometimes exhibit certain issues with the XY Picker mechanism, causing a failure to vend product. Symptoms may include:

- Product not transitioning into the delivery cup
- Product not transitioning into the delivery port
- XY Mechanism does not go to the right selection
- XY Mechanism behaving erratically

XY Picker Mechanism

ACTION:

If a DN3800/5800 is exhibiting any of the following issues with the XY Picker Mechanism, review the description of symptoms and troubleshooting steps to diagnose.

I. "Shelf Offset" Misalignment (Page 3)

Symptoms: If the "Shelf Offset" location setting is not set properly, the selected product may not successfully transition into product cup due to a misalignment of the cup plunger mechanism and the gate target. If the product cup moves into position, but the cup plunger does not impact the gate target to eject the product, the "Shelf Offset" settings may need adjustment. Refer to ***Page 3*** for detailed instruction on checking and adjusting the "Shelf Offset" location settings in the Service Mode.

II. "Hook Offset" Misalignment (Page 4)

Symptoms: If the "Hook Offset" location setting is not set properly, the product may not successfully transition from the product cup to the delivery port. If the product cup retrieves the product, then attempts unsuccessfully to deliver the product into the delivery port, the "Hook Offset" settings may need adjustment. Refer to ***Page 4*** for detailed instruction on checking and adjusting the "Hook Offset" location settings in the Service Mode.

III. Broken or Frayed XY Harnesses (Page 5)

Symptoms: If XY harness has become broken or frayed, it may lead to intermittent wiring connections that can cause unusual XY Picker behavior. If the XY picker mechanism is behaving erratically when trying to retrieve a selected product, refer to ***Page 5*** for detailed instructions on inspecting the XY harness.

IV. X Upper and Lower Carriages Binding or Off Track (Page 6)

Symptoms: If the XY Picker is having trouble delivering product, it may be a symptom of an issue with the X carriage assemblies. The X carriage assembly may be dislodged from the track, causing the XY Picker mechanism to bind on the rail. If the XY Picker is difficult to move by hand, it could be the result of a dislodged carriage or dirty carriage rails. Refer to ***Page 6*** for detailed instruction on inspection and replacement of the X Upper and Lower Carriages.

V. X Home, Y Home, and Plunger Home Switches (Page 14)

Symptom: If the X or Y Home switches are sticky with syrup or otherwise not functioning properly, it could cause issues with delivering product to the delivery cup or port. If a vender is having issues with delivering product to the cup or port, the X, Y, and Plunger Home switches should be inspected. Refer to ***Page 14*** for detailed instructions on inspecting those Home switches.

"Shelf Offset" Location Settings

Checking "Shelf Offset" Alignment:

1. Press the service switch on the controller (Blue button on the control board).
2. When the display reads "Error Codes," press and let go of the following keys in order: **1-5-1-5-1**. The VMC should enter the factory diagnostic mode and the display will read "BevMax Setup."
3. Press the **2** key to scroll to Position Test.
4. The left service door switch must be pulled to the out position to perform this test.
5. Press the **4** key one time to cause the VMC to enter Position Test. The display will show a set of numbers (###### # ###### #).
6. Press the **F** key to ensure XY assembly is in home position at the bottom left hand corner of the machine. Check VMC board in service area to ensure XY is at home position. If you look at the control board the green, amber (yellow), and red lights should be on at this time. If not, check the following: Green light is for home switch on bottom of port cup, amber (yellow) light is for home switch on left side of Y motor assembly, red light is for home switch for picker cup plunger.
7. With Left Door switch pulled and XY in the home position, test several locations.
8. To move to a specific location, select the shelf letter first (A,B,C,D,E) and allow the vend mechanism to come to a complete stop.
9. Next select a column number and allow the vend mechanism to come to a complete stop.
10. Next press the **0** key to cycle the plunger.
11. Top edge of the plunger should be striking the target between the two guidelines on the target face (see illustration below). <u>If Offset is hitting high or low then you need to change the offset.</u>

Adjusting Shelf Offset Alignment:

1. Press the **CLR** key to exit position test. Screen should read "Position Test."
2. Press left door switch to return to normal position.
3. Press the **3** key to scroll to Shelf Offset.
4. Press the **4** key one time to cause the VMC to enter Shelf Offset Menu.
   The VMC will enter the shelf adjust option and display the current shelf offset position as "Shelf Offset: ####, 3310 counts = 1 inch."
5. To adjust Shelf (Y) Offset (up and down), enter a new number.
   - For factory default setting, see label settings listed inside on service door on the side wall.
   - Work in increments of 50-100 to increase or decrease the offset.
   - It is recommended not to go below 900.
6. Press the ***** key to save and display will show "#### Offset Recorded."
7. Press the **CLR** to exit out of the menu. Screen should read "Shelf Offset."
8. Press the **2** key and return to "Position Test." Repeat steps to check adjustment.

Plunger Hitting Low

Correct

Plunger Hitting High

"Hook Offset" Location Settings

Checking Hook Offset Alignment:

1. Press the service switch on the controller (blue button on the control board).
2. When the display reads "Error Codes," press and let go of the following keys in order: **1-5-1- 5-1**. If the keys were actuated correctly, the VMC will enter the factory diagnostic mode and the display will read "BevMax Setup."
3. Press the **2** key to scroll to "Set Hook Offset." This sub-menu will be used to test the vend mechanism alignment to the port area and Adjust Hook Swipe "X" Offset (left to right).
4. Press the **4** key and the display will show "Hook Swipe ####, 3310 counts = 1 inch."
5. For factory default setting, see label settings listed inside on service door on the side wall.
6. Pull the top left (white) service door switch in the out position to power the XY motors.
7. Press the ***** key once. The XY will automatically run the Delivery Cup Assembly to the swipe position.
8. Review position of the cup sleeve to the discharge frame and determine if adjustment is necessary. Cup sleeve should be relatively centered to discharge frame lip. Cup sleeve should not impact light lens or wall prior to stopping. See illustration below.

Adjusting Hook Offset Alignment:

1. If adjustment is necessary, use the numbers on the keypad to change the coordinates. Change in increments of 50-100.
2. Press the ***** key to save the new setting and the Delivery Cup Assembly will automatically go to the "Home" position and then move back to the swipe position.
3. Repeat until correct position is achieved. Once position is achieved or if no adjustment is necessary. Press the ***** key to send the Delivery Cup Assembly back to the "Home" position and the programming will return to "Set Hook Offset."

Offset To Far Right

Correct

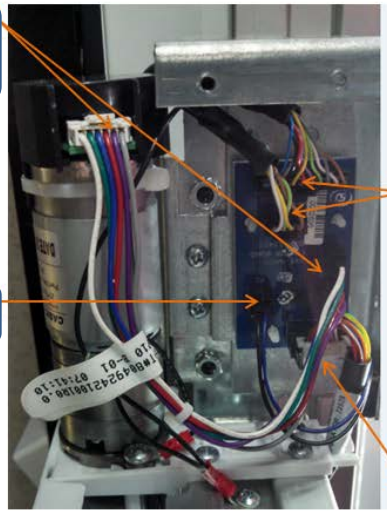
Offset To Far Left

Inspecting XY Harnesses

Inspection Steps:

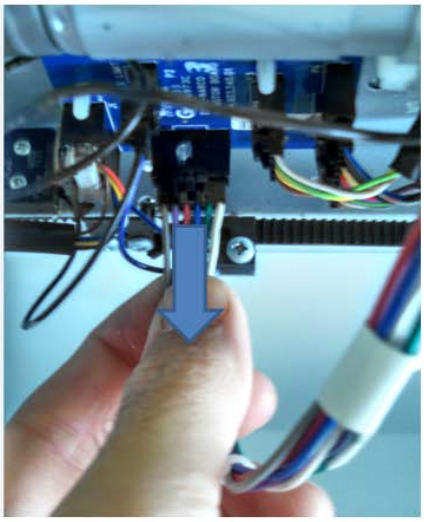
1. Remove the Y Motor Cover.

Remove Motor Cover Screws



Y Motor Board Connections

2. Inspect all harnesses.



Gently pull down on all connectors.

Ensure Connector Locks are Secure and Not Damaged

Ensure Wires Do not Pull Out Of Connectors

Inspect Wires For any Signs of Wear or Damage

3. Replace any worn or defective harnesses.

4. Replace Y-Board (CCP# 117074) if any connector locks are damaged.

X Upper and Lower Carriages

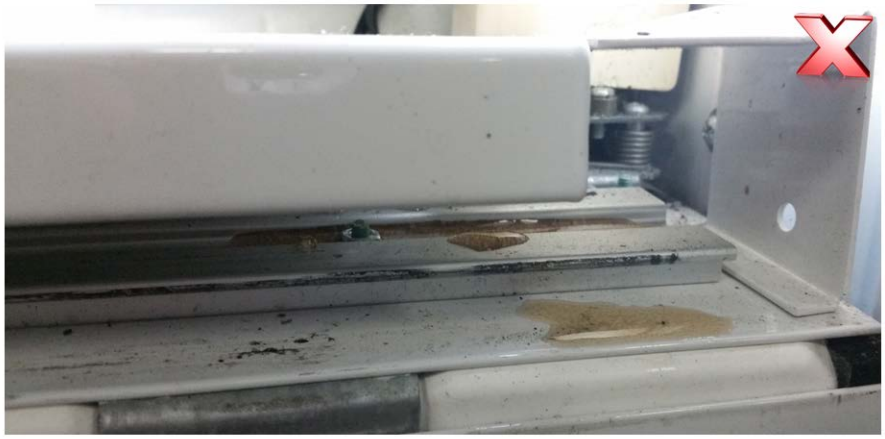
Upper/Lower Carriage Inspection:

1. Inspect both the upper and lower XY picker carriages per the illustrations below.
2. If the carriage is dislodged from the track or the carriage rollers appear loose or worn, replace the carriage assembly using the detailed instructions on the next page.

Move XY To Center and Visually Inspect upper and lower carriages.

All Carriage Wheels Should Be on XY Track.

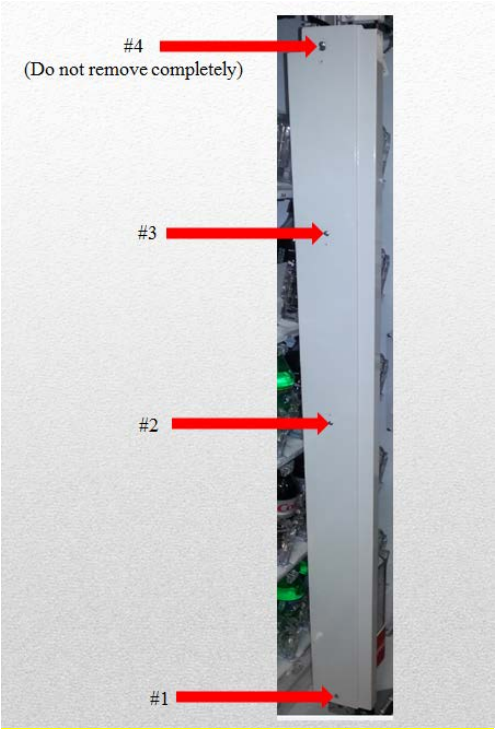
3. Inspect the carriage rails for any dirt or syrup buildup and clean if necessary.



Excessive dirt and syrup build up should be cleaned.

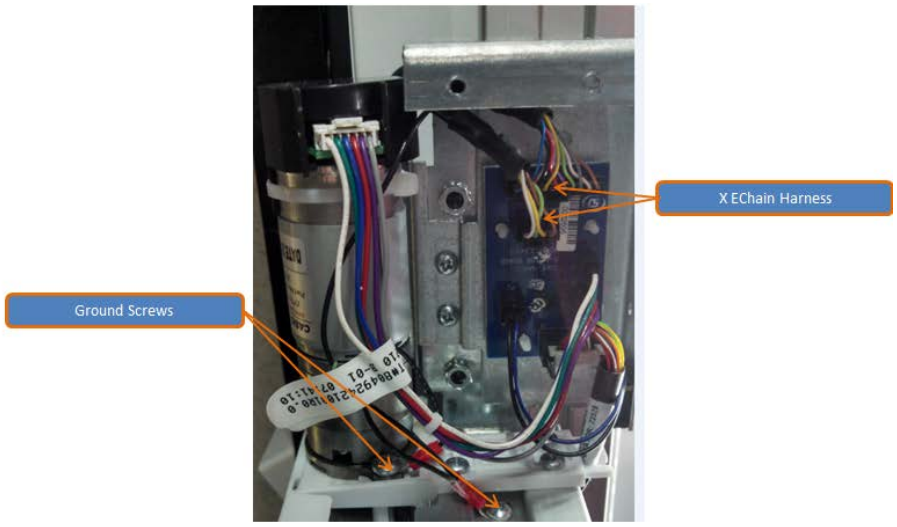
Upper/Lower Carriage Replacement:

1. Remove Y extrusion cover by removing screws 1-3 and loosening screw #4.



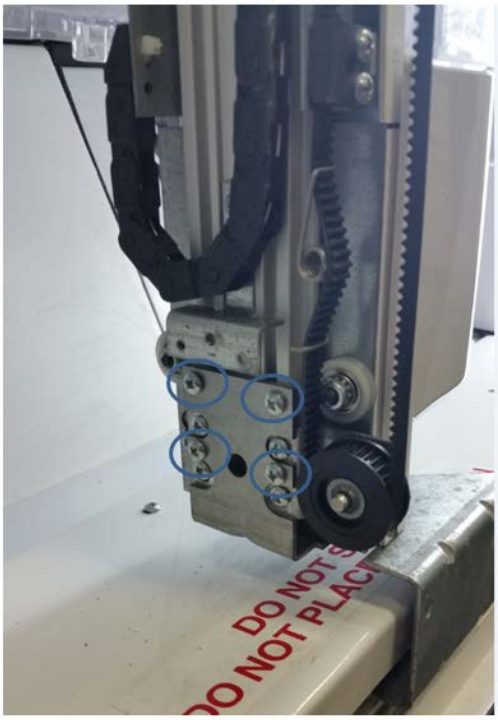
2. Remove Y Motor Cover and disconnect X E-Chain Harness.

Remove Motor Cover Screws



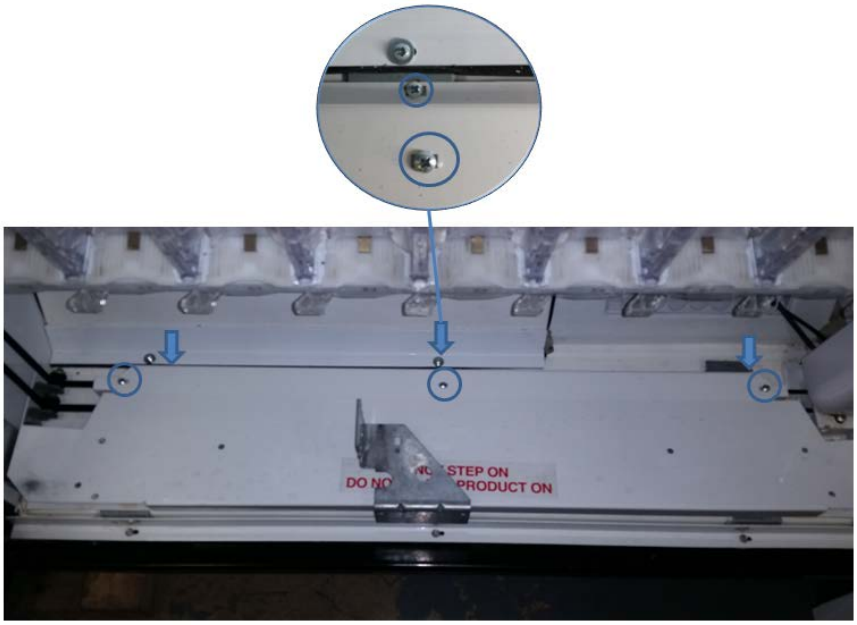
Disconnect X Echain and Ground Screws.

3. Disconnect the X Member from the upper and lower carriage by removing the 4 Phillips screws from the lower carriage. The Y Member Assembly can now be removed and set aside.

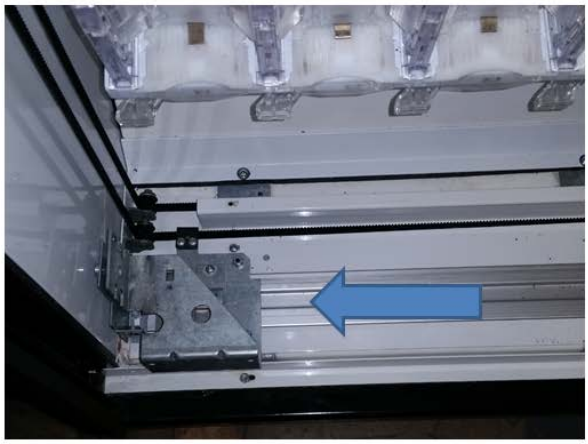


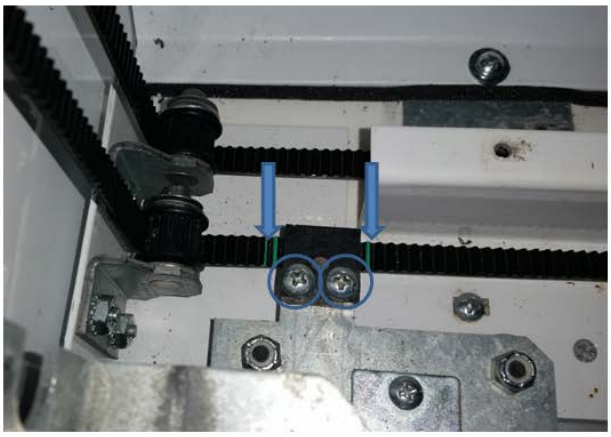
Lower Carriage Replacement:

4. If replacing the Lower Carriage, continue with steps 5-11. If only replacing the Upper Carriage, proceed to step 12.
5. Remove Lower X Belt Cover by removing the 6 Phillips head screws.



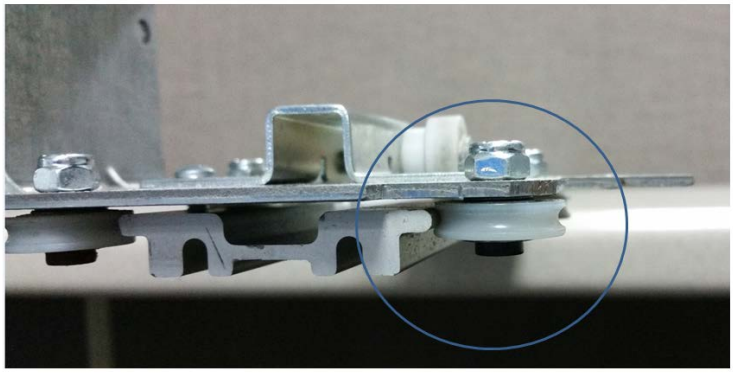
6. Move X carriage by hand all the way to the left of the cabinet. Mark both sides of the belt clamp with a paint pen. This will help when installing the new carriage. Then remove the two Phillips screws from belt clamp.





7. Using an 11/32” socket, remove the 3 nuts holding the X extrusion. Lift the extrusion up and remove the Lower X carriage assembly from the machine.

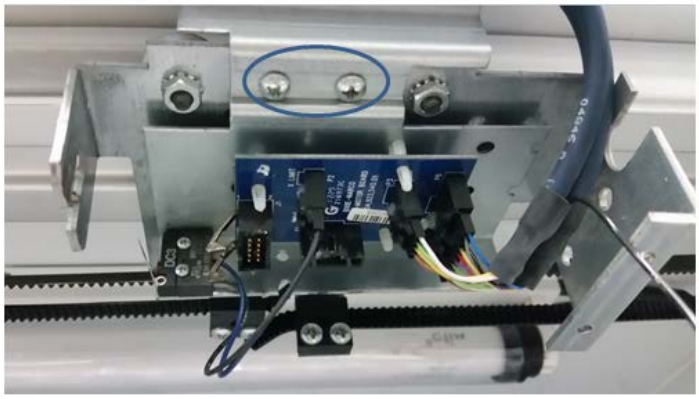
8. Remove old carriage by rolling off one end. Replace with new lower carriage (CCP# 129749). Ensure that all four rollers are on track before reassembling XY mechanism.

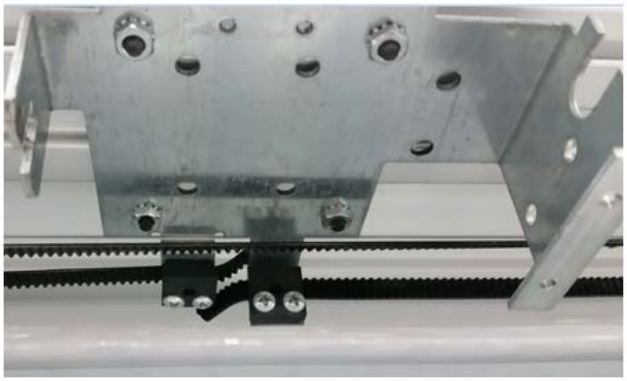


9. Reinstall Lower X extrusion into the cabinet with the new carriage attached.
10. Reinstall belt to belt clamp, utilizing alignment marks made during disassembly.
11. If also replacing the Upper Carriage, proceed to Step 12. If only replacing the Lower Carriage, reinstall the lower belt cover and Y Member Assembly. Be sure to reconnect the X E-Chain harness and ground screws.

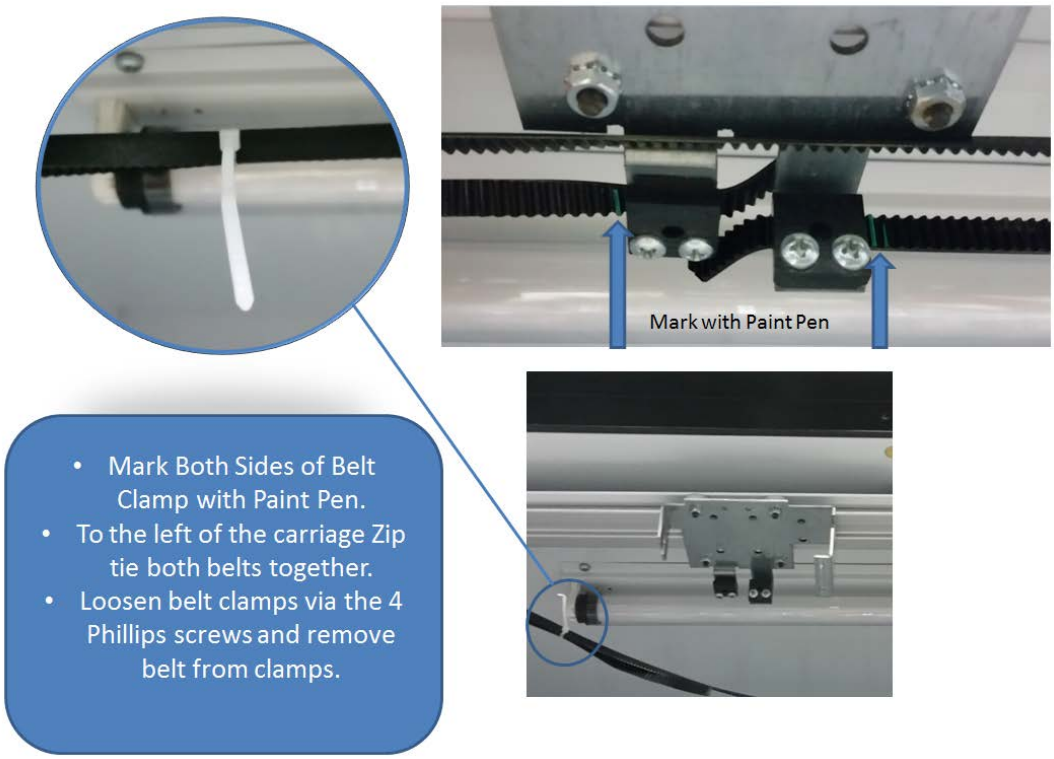
Upper Carriage Replacement:

12. Remove Y board assembly by taking out 2 Phillips screws.

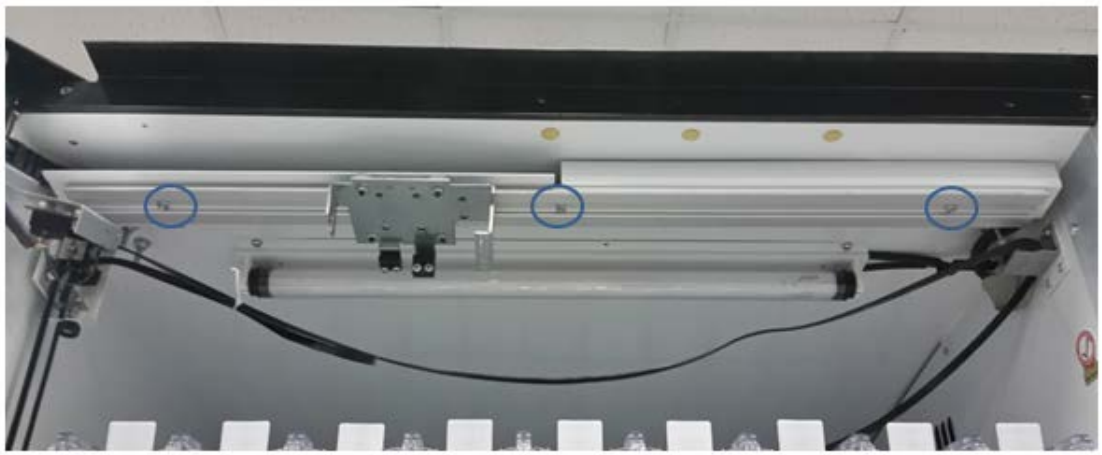




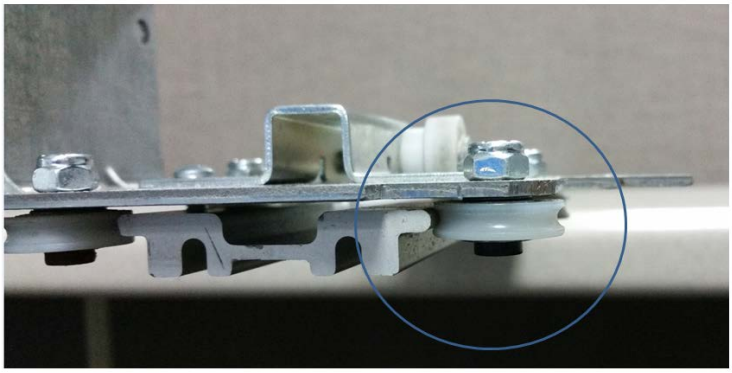
13. Mark belt with paint pen on either side of the belt clamps. To the left of the carriage, zip tie both belts together. This will aid in reassembly.
14. Remove belt from clamps by loosening the 4 Phillips screws.



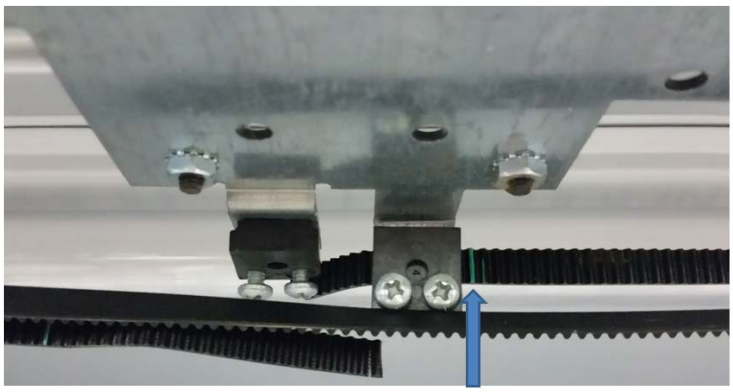
15. Using an 11/32" socket, remove the 3 nuts holding the X extrusion. Be sure to hold the X extrusion with one hand while removing the last nut. The extrusion will drop off as soon as the last nut is removed. Remove The X extrusion.



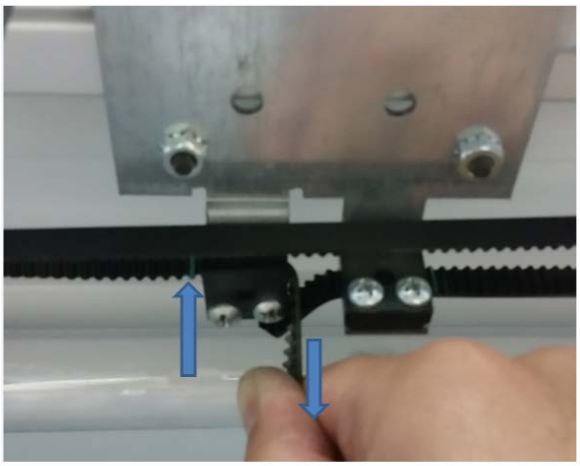
16. Remove carriage by sliding it to one side of extrusion. Replace new upper carriage (CCP# 129418) ensuring that all 4 wheels are on track.



17. Reinstall X extrusion into the cabinet with the new carriage attached.
18. Reinstall belt to belt clamps. Start with the right belt clamp first and utilize alignment marks made during disassembly.



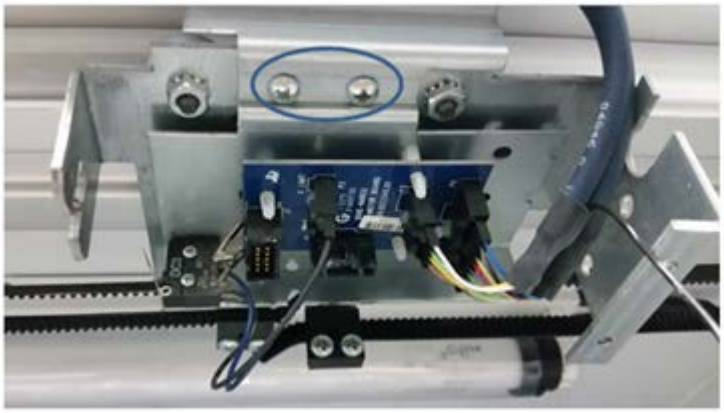
Reinstall belt into right belt clamp first. Align to mark previously made on belt and tighten belt clamp screws.



Reinstall belt into left belt clamp. Pull down on the belt end until previously made mark aligns to belt clamp. Tighten belt clamp screws.

19. Cut Zip Tie From Belt. Be careful not to cut the belt.

20. Reinstall the Y board assembly and reconnect the X E-Chain harness and ground screws.

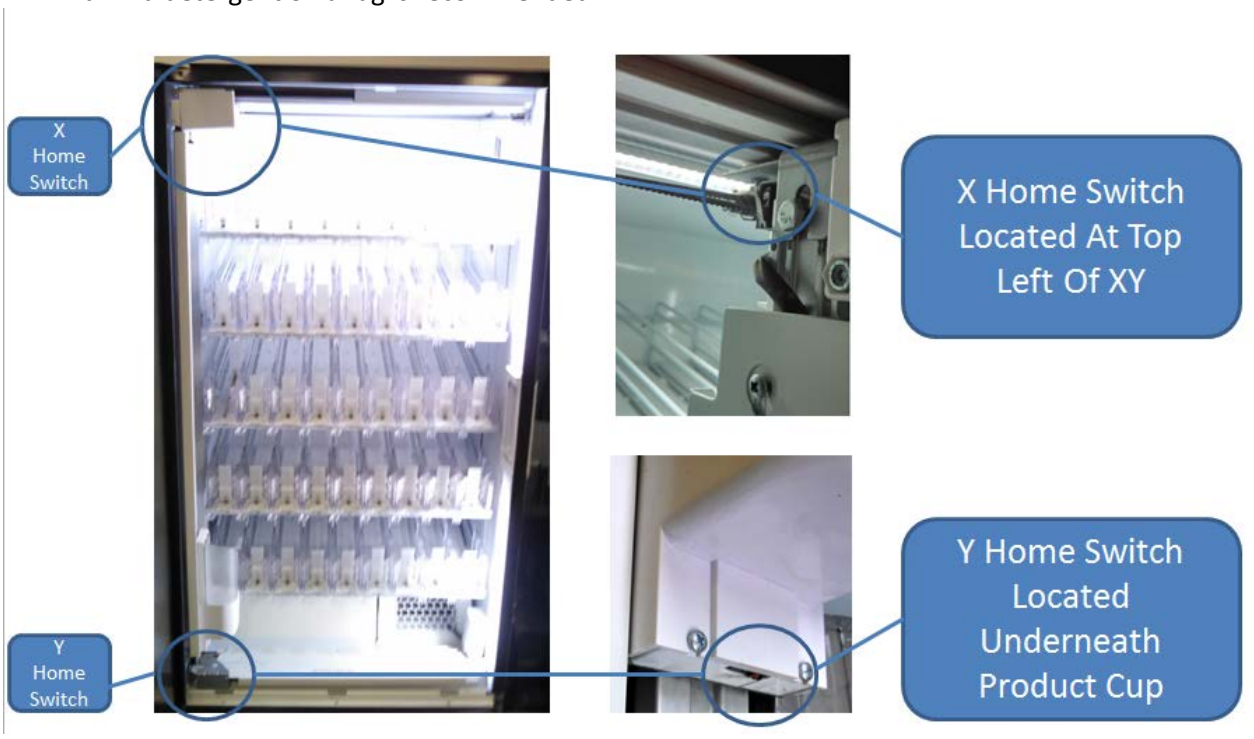


21. Reinstall the Y Member assembly and cover.

X Home, Y Home, and Plunger Home Switches

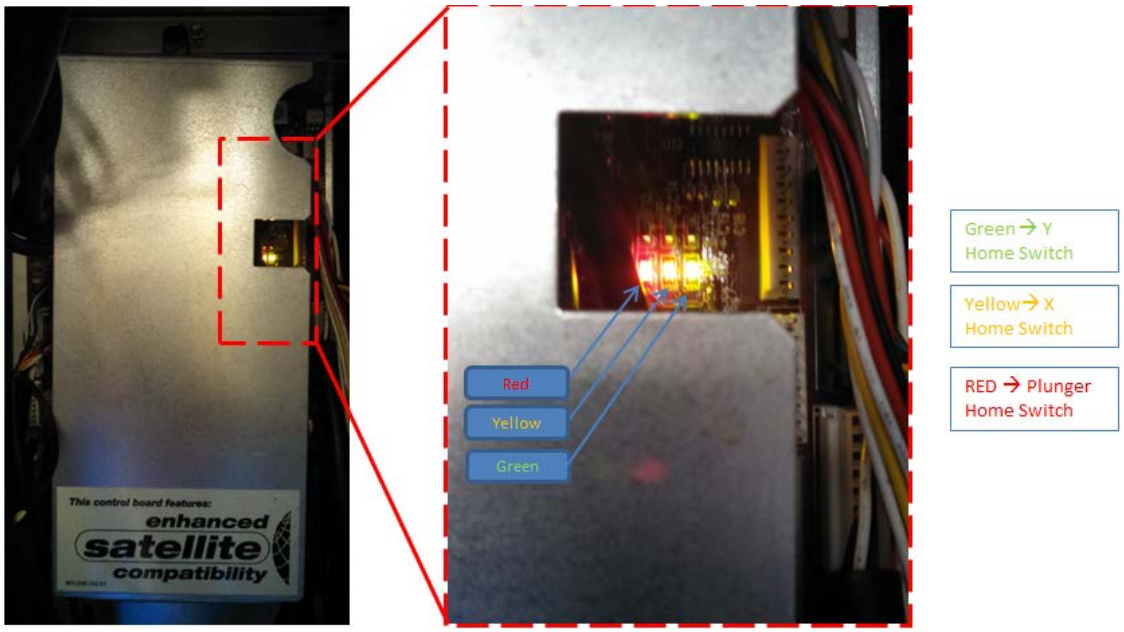
Inspecting X Home and Y Home Switches:

1. Inspect for syrup buildup around the X Home and Y Home Switches.
2. Clean all syrup/dirt from around home switches with an appropriate cleaner. Warm water with a mild detergent on a rag is recommended.



Checking the Functionality of X Home, Y Home, and Plunger Home Switches:

1. Press the service switch on the controller (blue button on the control board).
2. When the display reads "Error Codes," press and let go of the following keys in order: **1-5-1-5-1**. If the keys were actuated correctly, the VMC will enter the factory diagnostic mode and the display will read "BevMax Setup."
3. Press the **2** key to scroll to Position Test.
4. The left service door switch must be pulled to the out position to perform this test.
5. Press the **4** key one time to cause the VMC to enter Position Test.
6. Press the **F** Key to ensure XY assembly is in home position at the bottom left hand corner of the machine.
7. Check VMC board in service area to ensure XY is at home position. If you look at the control board the green, amber (yellow), and red lights should be on at this time. If not please check the following:
    a. Green light is for home switch on bottom of port cup.
    b. Amber (yellow) light is for home switch on left side of Y motor assembly.
    c. Red light is for home switch for picker cup plunger.



Testing Switch Responsiveness:

8. While still in Position Test with Left Door switch pulled, XY in the home position, and all lights on the board lit, press **A** while watching the VMC LED lights.
9. As soon as the cup moves off home, the Green light should go out. Any delay/flickering of the light indicates a possible issue with the switch or harness. If the Y Home switch is not functioning correctly, replace the Product Cup (CCP# 122667).
10. The Cup should end up at A1 location.
11. Press **0** to cycle the plunger. Ensure that it is hitting the A1 target location.

12. While watching the LED VMC lights, press **0** to cycle the plunger. The red light should immediately go off and then come back on. Failure to do so indicates an issue with the plunger home switch or harness. If the plunger home switch is not functioning correctly, replace the Product Cup (CCP# 122667).
13. With the cup at A1 location and the Yellow and Red Light on, press **5** while watching the LED VMC lights. The Yellow light should go off and the XY should move to column 5. Any delay/flickering of the light indicates a possible issue with the X Home switch or harness. Replace X Home Switch (CCP# 117772), if not functioning correctly.

FOR ADDITIONAL INFORMATION CONTACT:

PhoneFix® - Primary	Commercialization & Development	CCR Equipment Engineering	Coca-Cola Parts
	Glenn Dickerson	John Fudge	
(800) 817-2653	(404) 887-2752	(404) 676-8337	(800) 638-2653